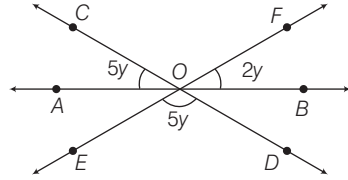


10. In the given figure, the value of y is

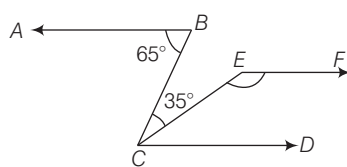


- (a) 25° (b) 35° (c) 15° (d) 40°

11. AB and CD are two parallel lines. PQ cuts AB and CD at E and F , respectively. EL is the bisector of $\angle FEB$. If $\angle LEB = 35^\circ$; then $\angle CFQ$ will be

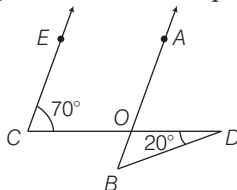
- (a) 110° (b) 85° (c) 70° (d) 95°

12. AB and CD are two parallel lines. The points B and C are joined such that $\angle ABC = 65^\circ$. A line CE is drawn making angle of 35° with the line CB , EF is drawn parallel to AB , as show in figure, then $\angle CEF$ is equal to



- (a) 160° (b) 155° (c) 150° (d) 145°

13. In the given figure, if $EC \parallel AB$, $\angle ECD = 70^\circ$, $\angle BDO = 20^\circ$, then $\angle OBD$ is equal to

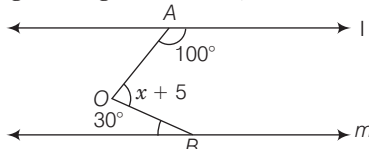


- (a) 70° (b) 60° (c) 50° (d) 20°

14. Two parallel lines AB and CD are intersected by a transversal line EF at M and N , respectively. The lines MP and NP are the bisectors of the interior angles BMN and DNM on the same side of the transversal. Then, $\angle MPN$ is equal to

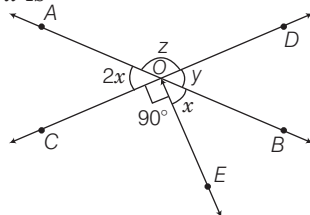
- (a) 90° (b) 45° (c) 135° (d) 60°

15. In the given figure if $l \parallel m$, then the value of x is



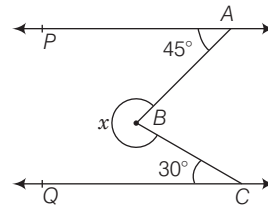
- (a) 105° (b) 100° (c) 110° (d) 115°

16. In the given figure, if $\angle COE = 90^\circ$, then the value of x is



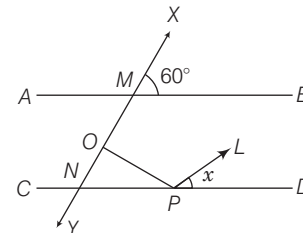
- (a) 120° (b) 60° (c) 45° (d) 30°

17. The value of x in the given figure is



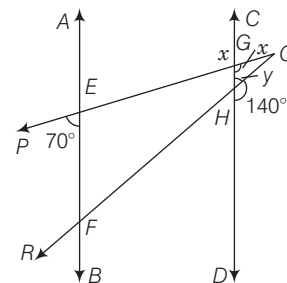
- (a) 75° (b) 185°
(c) 285° (d) 245°

18. In the given figure, $AB \parallel CD$, $\angle DPL = \frac{1}{2} \angle NPO$, and $OP \perp XY$ the value of x° is



- (a) 30° (b) 40° (c) 15° (d) 25°

19. In the given figure, $AB \parallel CD$ and they cut PQ and QR at E, F and G, H respectively. Then find the value of $x + y$



- (a) 120° (b) 130°
(c) 150° (d) 132°

20. Two parallel lines are cut by a transversal, then which of the following are true?

- I. Pair of alternate interior angles are congruent.
II. Pair of corresponding angles are congruent.
III. Pair of interior angles on the same side of the transversal are supplementary.

Select the correct answer using the codes given below

- (a) I, II and III are true (b) I and III are true
(c) I and II are true (d) II and III are true

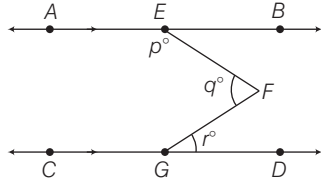
21. Consider the following statements related to three lines L_1, L_2 and L_3 in the same plane.

- I. If L_2 and L_3 are both parallel to L_1 , then they are parallel to each other.
II. If L_2 and L_3 are both perpendicular to L_1 then they are parallel to each other.
III. If there is acute angle between L_1 and L_3 , then L_2 is parallel to L_3 .

Which of the statement(s) given above is/are correct?

- (a) I and II (b) II and III
(c) All of these (d) None of these

- 22.** In the given figure, $AB \parallel CD$, then which one of the following is true?



- (a) $p + q - r = 180^\circ$ (b) $p + q + r = 180^\circ$
(c) $p - q + r = 180^\circ$ (d) $p + q - 2r = 180^\circ$

- 23.** LM is a straight line and O is a point on LM . Line ON is drawn not coinciding with OL or OM . If $\angle MON$ is one-third of $\angle LON$, then what is $\angle MON$ equal to?

- (a) 45° (b) 60°
(c) 75° (d) 80°

- 24.** Consider the following statements :

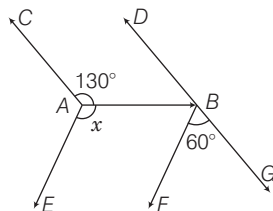
Two lines intersected by a transversal are parallel, if

- I. The pairs of corresponding angles are equal.
II. The interior angles on the same side of the transversal are supplementary.

Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 25.** In the given figure $AC \parallel BD$ and $AE \parallel BF$. What is the value of $\angle x$?



- (a) 130° (b) 110°
(c) 70° (d) 50°

- 26.** The necessary conditions for the line l and m to be parallel when these lines are intersected by a transversal line n is that

- I. Interior angles on the same side are equal.
II. Corresponding angles are equal.
III. Vertically opposite angles are equal.
IV. Alternate interior angles are equal.

Select the correct answer using the codes given below

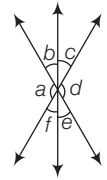
- (a) II and III (b) I, II and III (c) II and IV (d) I, II and IV

- 27.** In the given figure, which of the following statements must be true?

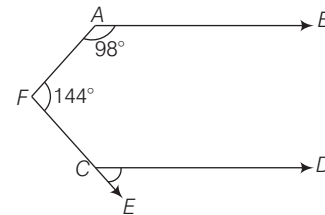
- I. $a + b = d + c$ II. $a + c + e = 180^\circ$
III. $b + f = c + e$ IV. $a + b + c = d + e + f$

Select the correct answer using the codes given below

- (a) Only I (b) I, II and III
(c) II, III and IV (d) All of these



- 28.** In the given figure, AB is parallel to CD . If $\angle BAF = 98^\circ$ and $\angle AFC = 144^\circ$, then $\angle ECD$ is equal to?



- (a) 62° (b) 64°
(c) 82° (d) 84°

- 29.** Consider the following statements :

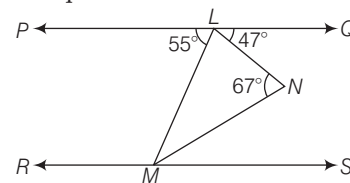
If two parallel lines are intersected by a transversal, then

- I. each pair of corresponding angles are equal.
II. each pair of alternate angle are unequal.

Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 30.** In the given figure, PQ is parallel to RS . Then $\angle NMS$ is equal to

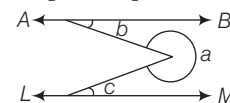


- (a) 20° (b) 23° (c) 27° (d) 47°

- 31.** The line segments AB and CD intersect at O . OF is the internal bisector of obtuse $\angle BOC$ and OE is the internal bisector of acute $\angle AOC$. If $\angle BOC = 130^\circ$, then what is the measure of $\angle FOE$?

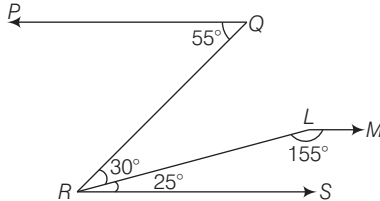
- (a) 90° (b) 110° (c) 115° (d) 120°

- 32.** In the given figure below, AB is parallel to LM . What is the angle a equal to?



- (a) $\pi + b + c$ (b) $2\pi - b + c$
(c) $2\pi - b - c$ (d) $2\pi + b - c$

33. In the given figure, PQ is parallel to RS . What is the angle between the lines PQ and LM ?



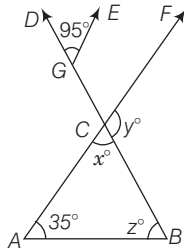
- (a) 175° (b) 177° (c) 179° (d) 180°

34. The length of a line segment AB is 2 units. It is divided into two parts at the point C such that $AC^2 = AB \times CB$. What is the length of CB ?

- (a) $3 + \sqrt{5}$ units (b) $3 - \sqrt{5}$ unit
(c) $2 - \sqrt{5}$ unit (d) $\sqrt{3}$ units

Directions (Q.Nos 35-36) Read the information and answer the questions.'

In the given figure, the lines CB and AC of a triangle ABC are extended to D and F , respectively and $CF \parallel GE$.



35. The sum of the value of x , y and z .
(a) 250° (b) 180° (c) 230° (d) 110°
36. The value of the angle which equals one-fifth of the supplement of z ?
(a) 50° (b) 26° (c) 10° (d) 130°

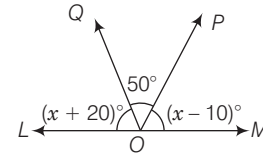
PREVIOUS YEARS' QUESTIONS

37. AB is a straight line. C is a point whose distance from AB is 3 cm. What is the number of points which are at a distance of 1 cm from AB and 5 cm from C ? **2012 I**
(a) 1 (b) 2 (c) 3 (d) 4
38. The ratio of two complementary angle is 1 : 5. What is the difference between the two angles? **2012 I**
(a) 60° (b) 90°
(c) 120° (d) Cannot be determined
39. Consider the following statements :
If two straight lines intersect, then
I. vertically opposite angles are equal.
II. vertically opposite angles are supplementary.
III. adjacent angles are complementary.

Which of the statement(s) given above is/are correct? **2012 I**

- (a) Only III (b) Only I (c) I and III (d) II and III

40. In the given figure, LOM is a straight line. What is the value of x° ? **2012 II**



- (a) 45° (b) 60° (c) 70° (d) 80°

41. If the arms of one angle are respectively parallel to the arms of another angle, then the two angles are **2013 I**

- (a) Neither equal nor supplementary
(b) not equal but supplementary
(c) equal but not supplementary
(d) Either equal or supplementary

42. The complement angle of 80° is **2015 I**

- (a) $\frac{18}{\pi}$ radian (b) $\frac{5\pi}{9}$ radian
(c) $\frac{\pi}{18}$ radian (d) $\frac{9}{5\pi}$ radian

43. Let OA, OB, OC and OD be rays in the anti-clockwise direction? Such that $\angle AOB = \angle COD = 100^\circ$, $\angle BOC = 82^\circ$ and $\angle AOD = 78^\circ$.

Consider the following statements

I. AOC and BOD are lines.

II. $\angle BOC$ and $\angle AOD$ are supplementary. **2015 I**

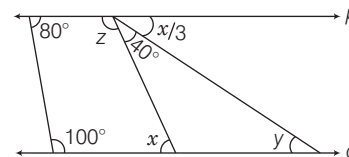
Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

44. If a transversal intersects four parallel straight lines, then the number of distinct values of the angles formed will be **2016 I**

- (a) 2 (b) 4 (c) 8 (d) 16

- 45.



In the figure given above, p and q are parallel lines. What are the values of the angles x , y and z ? **2016 I**

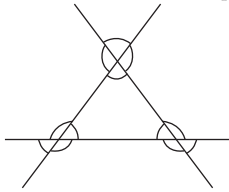
- (a) $x = 80^\circ$, $y = 40^\circ$, $z = 100^\circ$
(b) $x = 80^\circ$, $y = 50^\circ$, $z = 105^\circ$
(c) $x = 70^\circ$, $y = 40^\circ$, $z = 110^\circ$
(d) $x = 60^\circ$, $y = 20^\circ$, $z = 120^\circ$

ANSWERS

1	d	2	d	3	a	4	c	5	b	6	c	7	c	8	a	9	c	10	c
11	a	12	c	13	c	14	a	15	a	16	d	17	c	18	c	19	c	20	a
21	a	22	a	23	a	24	c	25	b	26	c	27	c	28	a	29	a	30	a
31	a	32	c	33	d	34	b	35	c	36	b	37	d	38	a	39	b	40	b
41	b	42	c	43	d	44	a	45	d										

HINTS AND SOLUTIONS

1. (d) We know that, when two lines intersect each other, it makes 4 angles. Since, the total number of pairs = 3



∴ Total number of angles = $3 \times 4 = 12$

2. (d) Let angle be x and its complement be $90^\circ - x$.

According to the question,

$$x = (90^\circ - x) + 14$$

$$\Rightarrow 2x = 104^\circ$$

$$\Rightarrow x = \frac{104^\circ}{2} = 52^\circ$$

3. (a) Let angle be x and its supplementary = $180^\circ - x$

$$\text{Then, } x = 2(180^\circ - x)$$

$$\Rightarrow 3x = 360^\circ$$

$$\Rightarrow x = 120^\circ$$

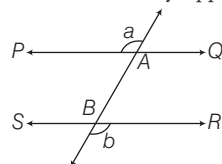
4. (c) The least number of straight lines for a bounded plane figure is 3.

5. (b) Given, angle = $\frac{3}{5}$ of right angle
 $= \frac{3}{5} \times 90^\circ = 3 \times 18^\circ = 54^\circ$

$$\text{Supplement of } 54^\circ = (180^\circ - 54^\circ)$$

$$= \text{An angle of measure } 126^\circ$$

6. (c) We have, $\angle QAB = \angle a$
 [vertically opposite angles]



$$\text{and } \angle QAB = \angle b$$

$$\Rightarrow \angle a = \angle b \quad [\text{corresponding angles}]$$

7. (c) Here, $\angle 1 = \angle 5$
 [corresponding angles]

$$\angle 7 = \angle 5 \quad (\text{vertical opposite})$$

$$\Rightarrow \angle 1 = \angle 7 \quad (\text{angle})$$

$$\text{So, } 5x - 20^\circ = 2x + 10^\circ$$

$$3x = 30^\circ \Rightarrow x = 10^\circ$$

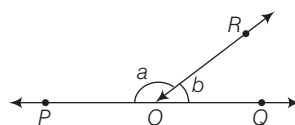
Hence,

$$\angle 7 = 2x + 10^\circ = 2(10^\circ) + 10^\circ = 30^\circ$$

8. (a) Complementary angle of $12^\circ 25' 40''$
 $= 90^\circ - 12^\circ 25' 40''$
 $= 89^\circ 59' 60'' - 12^\circ 25' 40''$
 $= (89 - 12)^\circ + (59' - 25') + (60'' - 40'')$
 $= 77^\circ + 34' + 20'' = 77^\circ 34' 20''$

9. (c) Since, $\angle POR$ and $\angle QOR$ is a linear pair.

$$\therefore \angle POR + \angle ROQ = 180^\circ$$



$$\Rightarrow a + b = 180^\circ \quad \dots (i)$$

$$\text{and } a - b = 80^\circ \quad \dots (ii)$$

On solving Eqs. (i) and (ii), we get

$$2a = 260^\circ \Rightarrow a = 130^\circ$$

$$\text{and } b = 180^\circ - 130^\circ = 50^\circ$$

$$\text{So, } a = 130^\circ, b = 50^\circ$$

10. (c) Since, AOB is a straight line.

$$\therefore \angle AOB = 180^\circ$$

$$\angle AOC + \angle COF + \angle FOB = 180^\circ$$

[straight line]

$$5y + 5y + 2y = 180^\circ$$

$$[\because \angle COF = \angle DOE, \text{ vertically opposite angles}]$$

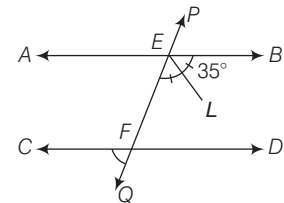
$$\Rightarrow 12y = 180^\circ$$

$$\Rightarrow y = \frac{180^\circ}{12} = 15^\circ$$

Hence, the value of y is 15° .

11. (a) Given, $\angle LEB = 35^\circ$

$$\angle FEB = 2 \times \angle LEB = 2 \times 35^\circ = 70^\circ$$



$$\Rightarrow \angle AEB = \angle AEF + \angle BEF = 180^\circ$$

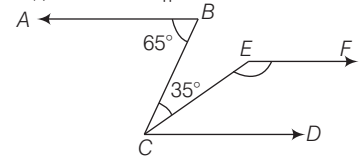
[straight line]

$$\Rightarrow \angle AEF = 180^\circ - 70^\circ = 110^\circ$$

$$\Rightarrow \angle CFQ = \angle AEF = 110^\circ$$

[corresponding angles]

12. (c) Since, $AB \parallel CD$



$$\therefore \angle ABC = \angle BCD = 65^\circ$$

[alternate angles]

$$\angle ECD = 65^\circ - \angle BCE$$

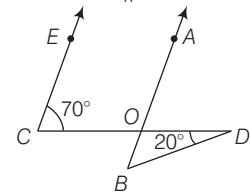
$$= 65^\circ - 35^\circ = 30^\circ$$

$$\angle CEF + \angle ECD = 180^\circ$$

[Angles on the same side of transversal]

$$\Rightarrow \angle CEF = 180^\circ - 30^\circ = 150^\circ$$

13. (c) Since, $EC \parallel AB$



$$\angle AOD = \angle ECO$$

[∵ corresponding angles]

$$\Rightarrow \angle AOD = 70^\circ$$

$$\text{So, } \angle BOD = 180^\circ - 70^\circ$$

$$= 110^\circ \quad [\text{linear point}]$$

Now, in $\triangle BOD$

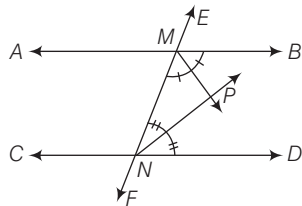
$$\angle OBD + \angle BOD + \angle ODB = 180^\circ$$

$$\Rightarrow \angle OBD = 180^\circ - (110 + 20)^\circ = 50^\circ$$

14. (a) Given, $\angle PMN = \frac{1}{2} \angle BMN$

$$\text{and } \angle PNM = \frac{1}{2} \angle DNM$$

As, $\angle BMN + \angle DNM = 180^\circ$
[angles on the same side of transversal]



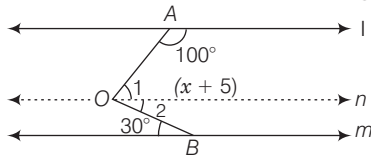
$\therefore$ In $\triangle PMN$, $\angle PMN + \angle PNM = 90^\circ$
 $\Rightarrow \angle MPN = 180^\circ - (\angle PMN + \angle PNM)$
[angle sum property]
 $\therefore \angle MPN = 180^\circ - 90^\circ = 90^\circ$

15. (a) Draw a line n passing through O and parallel to l and m

Since, $l \parallel n$, $\angle 1 + 100^\circ = 180^\circ$

[Sum of the interior angles on the same side of the transversal]
 $\angle 1 = 80^\circ$

Since, $n \parallel m$, $\angle 2 = 30^\circ$ [alternate angles]



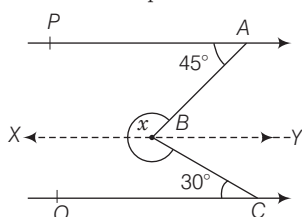
Now, $\angle AOB = \angle 1 + \angle 2$
 $= (80^\circ + 30^\circ) = 110^\circ$

But, $\angle AOB = (x + 5)^\circ = 110^\circ$
 $x = (110 - 5)^\circ = 105^\circ$

16. (d) Here, $\angle BOD = \angle AOC$
 $\therefore 2x = y$ [vertically opposite angles]
Now, COD is a straight line.

$\angle COD = 180^\circ$
 $\Rightarrow \angle COE + \angle EOB + \angle BOD = 180^\circ$
 $\Rightarrow 90^\circ + x + 2x = 180^\circ$
 $\Rightarrow 3x = 90^\circ \Rightarrow x = 30^\circ$
Hence, the value of x is 30° .

17. (c) From figure, $PA \parallel QC$
Draw a line XY parallel to PA and QC .



$\therefore PA \parallel XY$,
 $\Rightarrow \angle ABY = \angle PAB = 45^\circ$
[alternate angles]

and $\therefore XY \parallel QC$, $\angle QCB = \angle CBY = 30^\circ$
[alternate angles]

$\Rightarrow \angle ABC = \angle ABY + \angle CBY$
 $\Rightarrow \angle ABC = 45^\circ + 30^\circ = 75^\circ$
 $\therefore x = 360^\circ - \angle ABC = 360^\circ - 75^\circ$
 $\therefore x = 285^\circ$

18. (c) $\therefore AB \parallel CD$, $\angle ONP = \angle XMB = 60^\circ$
[corresponding angles]

$\angle OPN = 90^\circ - \angle ONP$
 $= 90^\circ - 60^\circ \Rightarrow \angle OPN = 30^\circ$

But, $\angle DPL = \frac{1}{2} \angle NPO = \frac{1}{2} 30^\circ$
 $\Rightarrow \angle DPL = 15^\circ \Rightarrow x = 15^\circ$

19. (c) Since, $AB \parallel CD$ and PQ is transversal
 $\angle PEF = \angle EGH$ [corresponding angles]
 $\Rightarrow \angle EGH = 70^\circ$ [$\because \angle PEF = 70^\circ$]
Now, $\angle EGH + \angle HGQ = 180^\circ$

[linear pair]

$\Rightarrow \angle HGQ = 180^\circ - 70^\circ = 110^\circ$

Also, $\angle DHQ + \angle GHQ = 180^\circ$

[linear pair]

$\angle GHQ = 180^\circ - 140^\circ = 40^\circ$

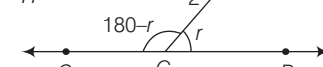
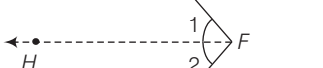
$\therefore x + y = 110^\circ + 40^\circ = 150^\circ$

20. (a) All the three statements are true.

21. (a) Only statements I and II are true.

22. (a) Draw $FH \parallel AB \parallel CD$
(sum of interior angles)

$\therefore \angle 1 + p = 180^\circ$... (i)



$\angle 2 + 180^\circ - r = 180^\circ$... (ii)

(sum of interior angle)

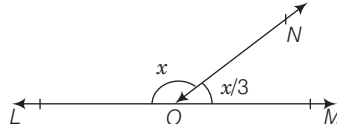
On adding Eqs. (i) and (ii), we get

$\angle 1 + \angle 2 + p + 180^\circ - r = 360^\circ$

$\Rightarrow p + q - r = 180^\circ$ [$\because \angle 1 + \angle 2 = q^\circ$]

23. (a) Given that, $\angle MON = \frac{1}{3} \angle LON$

Let $\angle LON = x$, then, $\angle MON = \frac{x}{3}$



We know that,

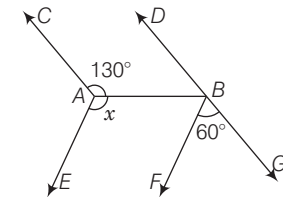
$\angle LON + \angle MON = 180^\circ$ [linear pair]

$\Rightarrow x + \frac{x}{3} = 180^\circ \Rightarrow x = \frac{180^\circ \times 3}{4} = 135^\circ$

Thus, $\angle MON = \frac{x}{3} = \frac{135^\circ}{3} = 45^\circ$

24. (c) Both the statements I and II are correct.

25. (b) Since, $AC \parallel BD$



$\therefore \angle DBA = 180^\circ - 130^\circ = 50^\circ$

[interior angle]

[$\because \angle BAC = 130^\circ$]

DBG is straight line.

$\therefore \angle DBA + \angle ABF + \angle FBG = 180^\circ$

[linear pairs]

$\Rightarrow 50^\circ + \angle ABF + 60^\circ = 180^\circ$

$\Rightarrow \angle ABF = 70^\circ$

Since, $AE \parallel BF$

$\therefore x = 180^\circ - \angle ABF$

$= 180^\circ - 70^\circ = 110^\circ$

26. (c) When two lines are parallel intersected by a transversal then corresponding as well as alternate interior angles are equal.

Hence, the statement II and IV are correct.

27. (c) We have, $a = d$, $b = e$ and $c = f$
[vertically opposite angles]

and $a + b + c = d + e + f = 180^\circ$

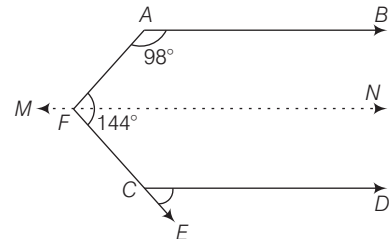
II. $a + b + c = 180^\circ \Rightarrow a + c + e = 180^\circ$

III. $b + f = c + e$

IV. $a + b + c = d + e + f$

Hence, statements II, III and IV are true.

28. (a) Draw a line MN parallel to AB and CD .



$\Rightarrow \angle AFN = 180^\circ - 98^\circ = 82^\circ$

(sum of interior angles)

$\angle CFN = 144^\circ - 82^\circ = 62^\circ$

and $\angle ECD = \angle CFN = 62^\circ$

[corresponding angles]

29. (a) If two parallel lines are intersected by a transversal, then each pair of corresponding angles and of alternate angles are equal.

Therefore, statement I is correct.